ClaimLens — Dataset Details

Dataset Reference & Preprocessing Specification

1. Dataset Overview

- **Dataset Name:** Free Synthetic Insurance Claims 100M
- **Primary Hugging Face Source:** ziadatalabs/FreeInsuranceClaims100M
- **Alternate Source:** Kaggle Insurance Claims Fraud Data
- **Format:** Apache Parquet (100 row groups, ~2.35 GB compressed) & sample CSV
- **Total Records:** 100,000,000 synthetic property & casualty insurance records
- **Coverage Lines:** Auto, Home, Renters, Property, and Business Insurance

2. Schema & 13 Core Fields Specification

Field Name	Data Type	Description	Role in Solution
claim_id	String	Unique claim identifier (e.g., CLM-0000000001)	Document ID in Vector DB
policy_id	String	Unique policy contract identifier	Policyholder linking
claim_type	String	Coverage line (Auto, Home, Renters, Business)	Metadata filter & ML feature
state	String	2-letter US State postal code	Geographic risk & filtering
policyholder_tenure_years	Float	Policyholder tenure duration with carrier	Fraud risk & triage scoring
previous_claims_count	Integer	Number of previous claims filed	Loss frequency feature
incident_date	Date	Date when the loss/damage occurred	Timeline validation
claim_filed_date	Date	Date claim was officially submitted	Filing latency indicator
claim_amount	Float	Total claimed loss amount in USD	Regression target & exposure
deductible	Float	Applicable policy deductible in USD	Net payout calculation
claim_status	String	Open, Approved, Closed, Denied, Under Invest.	Historical outcome precedent
days_to_resolution	Float	Turnaround days from filing to closing	Regression target (duration)
is_fraud_flagged_ground_truth	Boolean	Ground truth indicator for suspected fraud	Classification target (Fraud)

3. Data Preprocessing & Sampling Pipeline

- **Parquet Chunk Streaming:** PyArrow streams the first row group (row group 0) directly from Hugging Face, extracting the first 10,000 records sequentially for machine learning training, with 2,000 candidates indexed into the active in-memory vector store for 3.08 ms median matrix cosine similarity calculations.
- **Sampling Specification:** The 10,000 records represent a sequential partition sample from row group 0 (non-stratified single-group extraction), chosen to avoid multi-gigabyte disk download constraints while maintaining schema fidelity.
- **Handling Missing Values:** Open/pending claims have null days_to_resolution; these are preserved for open status queries and imputed during ML training on closed claims.
- **Text Narrative Synthesis:** Each row is structured into a natural language description chunk capturing all 13 attributes for dense embedding indexing.
- **Feature Engineering:** Derived filing_delay_days (difference between filing date and incident date) and claim_to_deductible_ratio for enhanced fraud risk prediction.